Implement the DELETE Endpoint for Removing an Existing Product

Explore how to implement the DELETE method in a Python REST API using Flask. Learn to map URI path parameters to function parameters, remove items from a product list, and return appropriate HTTP status codes to confirm resource deletion.

The equivalent of the Delete CRUD operation in REST is the DELETE method. It can be used in REST API to indicate that a particular resource in a collection needs to be removed or marked as not processable. If one looks at the URI for DELETE, it will look the same as that for GET. Hence, from a URI perspective, DELETE and GET use the same template.

In Flask, the path parameter of the URI gets mapped to the parameter of the method. The product ID passed as the path parameter can be used to delete the corresponding product from the list. In this section, let's see how to provide a DELETE REST endpoint.

Steps

1. Open `server/api/products_api.py`.
2. Add a new method that accepts a parameter named `product_id`. Name the method `delete_product`.

Ask

```
def delete_product(product_id):
```

3. Add the following code to the `delete_product` method.

```
for product in products:
    if product['id'] == product_id:
        products.remove(product)
        break
response = make_response('', 204)
response.mimetype = 'application/json'
return response
```

4. Decorate the method with the `route()` decorator of `products_blueprint`. Pass the URI that will represent the endpoint and the supported HTTP method, which is `DELETE`. Add the path parameter named `product_id` to the URI.

```
@products_blueprint.route('/products/<product_id>', methods=['DELETE'])
```

5. Run the application.
6. Hit the following URL using your preferred tool. When testing on the Educative platform, please replace `http://localhost` with the URL provided in the SPA widget.

```
$ curl -X DELETE http://localhost:5000/api/products/1
```

7. This should result in the following output.

```
< HTTP/1.1 204
```

How it works

The core of the logic is in the `delete_product` method and its `route()` decorator. The URI passed to the `route()` decorator can have a route parameter. This parameter maps with the path parameter of the URI, and it's passed as the parameter of the method that `route()` is decorating.

Ask

Request /response for DELETE API for removing an existing product

In our case, the `product_id` passed as the path parameter is used to find the corresponding product within the list and remove it from the list. Once the `remove` operation is successful, `204 N0 CONTENT` status without any body is returned as the response.

Try it yourself

- The widget below shows the code created during the previous lesson.
- Follow the steps above to complete the code for this lesson.
- Click “Run” in the widget below to start the service. The app URL for accessing the service is mentioned at the bottom of the widget.
- Wait for the service to start before accessing the URL.
- Append `/1` to the app URL to send the DELETE request.

> **Note:** Use REST API clients, such as Postman and Insomnia, or command line tools like curl to send the request.

Ask

- Send a GET request by appending `/1` to the app URL To verify that the deletion was successful.

> **Note:** Access the URL using any web browser, REST API clients, such as Postman and Insomnia, or command line tools like curl.

Ask

Run the code above, and access the URL once the service starts, to view the output

Solution

If stuck, refer to the final code for help.

Please note that the file below is read-only. The changes from the previous lesson are highlighted.

products_api.py Python 3.8

```
from flask import (request, jsonify, Blueprint, make_response, url_for)
from .data_store import products

products_blueprint = Blueprint('products_api', __name__, url_prefix='/api


@products_blueprint.route('/products', methods=['GET'])
def list_products():
    return jsonify(products)


@products_blueprint.route('/products/<product_id>', methods=['GET'])
def get_product(product_id):
```

Solution code for this lesson

Mark As Completed

Ask